

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	K Suthesika	Reg. No.:	192511070
Section / Batch:	BE CSE	Date:	24/07/2026

Code of Conduct

I, K Suthesika (192511070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Suthesika

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SET - 1

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack - Balancing Symbols
5 marks



Q6

Singly Linked List
5 marks



Q7

Stack
5 marks



Q8

Linked List
5 marks



Q9

Linked List
5 marks



Q10

Queue
5 marks



10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int n, int key) {  
4     int low = 0, high = n - 1;  
5  
6     while (low <= high) {  
7         int mid = (low + high) / 2;  
8  
9         if (arr[mid] == key)  
10            return mid;  
11        else if (arr[mid] < key)  
12            low = mid + 1;  
13        else  
14            high = mid - 1;  
15    }  
16    return -1;  
17 }  
18  
19 int main() {  
20     int arr[] = {10,20,30,40,50};  
21     int n = 5;  
22     int key = 30;  
23  
24     int result = binarySearch(arr,n,key);  
25     if (result != -1)  
26         printf("Element found at index %d\n", result);  
27     else  
28         printf("Element not found\n");  
29     return 0;  
30 }
```

Test Input

stdin input (optional)

Output

Element found at index 2

⏸ Pending manual review

✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 int pop() {  
9     if (top == -1) {  
10         printf("Stack Underflow\n");  
11         return -1;  
12     }  
13     return stack[top--];  
14 }  
15  
16 int main() {  
17     stack[++top]=10;  
18     stack[++top]=20;  
19     stack[++top]=30;  
20     printf("Deleted element: %d\n",pop());  
21     printf("Deleted element: %d\n",pop());  
22     printf("Deleted element: %d\n",pop());  
23     printf("Deleted element: %d\n",pop());  
24  
25     return 0;  
26 }
```

Test Input

stdin input (optional)

Output

```
Deleted element: 20  
Deleted element: 20  
Deleted element: 10  
Stack Underflow  
Deleted element: -1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

Q1 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
```

```
int top;
```

```
void init() {
```

```
top = 0;
```

```
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = -1;
8 }
9
10 int main() {
11     init();
12     printf("Top = %d\n", top);
13     return 0;
14 }
```

Test Input

stdin input (optional)

Output

Top = -1

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 char stack[MAX];  
6 char postfix[MAX];  
7 int top = -1;  
8 int j = 0;  
9  
10 void push(char ch) {  
11     stack[++top] = ch;  
12 }  
13  
14 char pop() {  
15     if (top == -1)  
16         return '\0';  
17     return stack[top--];  
18 }  
19  
20 int main() {  
21     char ch = '(';  
22  
23     push('(');  
24     push('+');  
25     push('*');  
26  
27     if (ch == ')') {  
28         while (top != -1 && stack[top] != '(') {  
29             postfix[j++] = pop();  
30         }  
31         if (top != -1)  
32             pop();  
33     }  
34     postfix[j] = '\0';  
35  
36     printf("Postfix: %s\n", postfix);  
37  
38     return 0;  
39 }
```

Test Input

stdin input (optional)

Output

Postfix: ++

 Pending manual review Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if(ch == '[' && stack[top] == ']')  
(ch == '[' && stack[top] == '[')  
pop();  
} else {  
return 0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 char stack[MAX];  
6 int top = -1;  
7  
8 void push(char ch) {  
9     stack[++top] = ch;  
10 }  
11 char pop() {  
12     if (top == -1)  
13         return '\0';  
14     return stack[top--];  
15 }  
16  
17 int main() {  
18     char exps[] = "{[()]}";  
19     int i = 0;  
20  
21     while (exps[i] != '\0') {  
22         char ch = exps[i];  
23  
24         if (ch == '[' || ch == '{' || ch == '(') {  
25             push(ch);  
26         }  
27         else if (ch == '}' || ch == ']' || ch == ')') {  
28             if (top == -1) {  
29                 printf("Not Balanced\n");  
30                 return 0;  
31             }  
32             if ((ch == '}' && stack[top] == '(') ||  
33                 (ch == ']' && stack[top] == '[') ||  
34                 (ch == ')' && stack[top] == '{')) {  
35                 pop();  
36             }  
37             else {  
38                 printf("Not Balanced\n");  
39                 return 0;  
40             }  
41         }  
42  
43         i++;  
44     }  
45     if (top == -1)  
46         printf("Balanced\n");  
47     else  
48         printf("Not Balanced\n");  
49  
50     return 0;  
51 }
```

Submitted

Test Input

stdin input (optional)

Output

Balanced

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7
8 int main() {
9     struct Node n1,n2,n3;
10    struct Node *head, *temp;
11
12    n1.data = 10;
13    n2.data = 20;
14    n3.data = 30;
15
16    n1.next = &n2;
17    n2.next = &n3;
18    n3.next = NULL;
19
20    head = &n1;
21
22    temp = head;
23    while (temp != NULL) {
24        printf("%d ", temp->data);
25        temp = temp->next;
26    }
27    return 0;
28 }
```

Test Input

stdin input (optional)

Output

10 20 30

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

@lse:

return stack[top--]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if(top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int pop()
19 {
20     if(top == -1)
21     {
22         printf("Stack Underflow\n");
23         return -1;
24     }
25     return stack[top--];
26 }
27
28 int main()
29 {
30     int n,i,x;
31
32     scanf("%d", &n);
33
34     for(i=0; i<n; i++)
35     {
36         scanf("%d", &x);
37         push(x);
38     }
39     printf("%d\n", pop());
40
41     return 0;
42 }
```

Test Input

3
10
20
30

Output

30

⏸ Pending manual review

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Doubly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *prev = NULL, *current, *next;
11
12     struct Node *n1 = (struct Node *)malloc(sizeof(struct Node));
13     struct Node *n2 = (struct Node *)malloc(sizeof(struct Node));
14     struct Node *n3 = (struct Node *)malloc(sizeof(struct Node));
15
16     n1->data = 10;
17     n2->data = 20;
18     n3->data = 30;
19
20     n1->next = n2;
21     n2->next = n3;
22     n3->next = NULL;
23
24     head = n1;
25     current = head;
26
27     while (current != NULL) {
28         next = current->next;
29         current->next = prev;
30         prev = current;
31         current = next;
32     }
33     head = prev;
34
35     current = head;
36     while (current != NULL) {
37         printf("%d ", current->data);
38         current = current->next;
39     }
40     return 0;
41 }
```

Test Input

stdin input (optional)



Output

20 20 10

🔍 Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *new;
12
13     new = (struct Node *)malloc(sizeof(struct Node));
14
15     new->data = 10;
16
17     new->next = head;
18     head = new;
19
20     struct Node *temp = head;
21     while (temp != NULL) {
22         printf("%d ", temp->data);
23         temp = temp->next;
24     }
25     return 0;
26 }
```

Submitted

Test Input

stdin input (optional)



Output

10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int Q[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int x) {  
9     if (rear == MAX - 1) {  
10         printf("Queue Overflow\n");  
11         return;  
12     }  
13     if (front == -1)  
14         front = 0;  
15     Q[++rear] = x;  
16 }  
17  
18  
19 int dequeue() {  
20     if (front == -1) {  
21         printf("Queue Underflow\n");  
22         return -1;  
23     }  
24     int item = Q[front++];  
25  
26     if (front > rear) {  
27         front = rear = -1;  
28     }  
29     return item;  
30 }  
31  
32 int main() {  
33     int n, i, w;  
34  
35     scanf("%d", &n);  
36  
37     for (i=0; i<n; i++) {  
38         scanf("%d", &w);  
39         enqueue(w);  
40     }  
41     printf("%d\n", dequeue());  
42     return 0;  
43 }
```

Test Input

3
10
20
30

Output

10

Pending manual review

Submitted

SET - 2

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks



Q2

Circular Queue
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Doubly Linked List
5 marks



Q6

Queue
5 marks



Q7

Queue - Deque
5 marks



Q8

Binary Search
5 marks



Q9

Circular Queue
5 marks



Q10

Stack - Postfix Expression
5 marks



10/10 answered

Q1 Stack

5 marks

Point out the error?

```
PEEK(stack);  
return stack[top+1]
```

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 void push(int value)  
9 {  
10     if(top == MAX - 1)  
11     {  
12         printf("Stack Overflow\n");  
13         return;  
14     }  
15     stack[++top] = value;  
16 }  
17  
18 int peek()  
19 {  
20     if(top == -1)  
21     {  
22         printf("Stack Underflow\n");  
23         return -1;  
24     }  
25     return stack[top];  
26 }  
27  
28 int main()  
29 {  
30     int n, i, x;  
31  
32     scanf("%d", &n);  
33     for(i=0; i<n; i++)  
34     {  
35         scanf("%d", &x);  
36         push(x);  
37     }  
38     printf("%d", peek());  
39  
40     return 0;  
41 }
```

Test Input

3
10
20
30

Output

30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1;
7 int rear = -1;
8
9 void enqueue(int vehicle)
10 {
11     if ((rear+1)% MAX == front)
12     {
13         printf("Queue Full\n");
14         return;
15     }
16     if (front == -1)
17         front = 0;
18     rear = (rear+1) % MAX;
19     queue[rear] = vehicle;
20 }
21 int main()
22 {
23     int n, i, x;
24
25     scanf("%d", &n);
26
27     for (i=0; i<n; i++)
28     {
29         scanf("%d", &x);
30         enqueue(x);
31     }
32     printf("Front = %d\n", queue[front]);
33     printf("Rear = %d\n", queue[rear]);
34
35     return 0;
36 }
```

Test Input

```
3
101
102
103
```

Output

```
Front = 101
Rear = 103
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if(top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int main()
19 {
20     int n, i, x;
21
22     scanf("%d", &n);
23
24     for (i=0; i<n; i++)
25     {
26         scanf("%d", &x);
27         push(x);
28     }
29     if(top != -1)
30         printf("%d\n", stack[top]);
31     return 0;
32 }
```

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Test Input

3
10
20
30

Output

30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if(top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int pop()
19 {
20     if (top == -1)
21     {
22         printf("Stack Underflow\n");
23         return -1;
24     }
25     return stack[top--];
26 }
27
28 int main()
29 {
30     int n, i, x;
31     scanf ("%d", &n);
32     for(i=0;i<n;i++)
33     {
34         scanf("%d", &x);
35         push(x);
36     }
37     printf("%d\n", pop());
38     return 0;
39 }
```

Test Input

3
10
20
30

Output

30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - postfix Expression
5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct node  
5 {  
6     int data;  
7     struct node *prev;  
8     struct node *next;  
9 };  
10 void deletenode(struct node **head, struct node *del)  
11 {  
12     if (del == NULL)  
13         return;  
14     if (*head == del)  
15         *head = del->next;  
16     if (del->prev != NULL)  
17         del->prev->next = del->next;  
18     if (del->next != NULL)  
19         del->next->prev = del->prev;  
20  
21     free(del);  
22 }  
23 void printlist(struct node *head)  
24 {  
25     while (head != NULL)  
26     {  
27         printf("%d ", head->data);  
28         head = head->next;  
29     }  
30 }  
31 int main()  
32 {  
33     struct node *head;  
34     struct node *n1 = (struct node *)malloc(sizeof(struct node));  
35     struct node *n2 = (struct node *)malloc(sizeof(struct node));  
36     struct node *n3 = (struct node *)malloc(sizeof(struct node));  
37  
38     n1->data=10;  
39     n2->data=20;  
40     n3->data=30;  
41  
42     n1->prev=NULL;  
43     n1->next=n2;  
44  
45     n2->prev=n1;  
46     n2->next=n3;  
47  
48     n3->prev=n2;  
49     n3->next=NULL;  
50  
51     head=n1;  
52  
53     deletenode(&head, n2);  
54  
55     printlist(head);  
56     return 0;  
57 }
```

Test Input

stdin input (optional)

Output

10 30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 void enqueue(int data)  
10 {  
11     if(rear == MAX - 1)  
12     {  
13         printf("Queue Overflow\n");  
14         return;  
15     }  
16     if (front == -1)  
17         front = 0;  
18     queue[++rear] = data;  
19 }  
20 int main()  
21 {  
22     int n, i, x;  
23     scanf("%d", &n);  
24     for(i=0; i<n; i++)  
25     {  
26         scanf("%d", &x);  
27         enqueue(x);  
28     }  
29     printf("Front = %d\n", queue[front]);  
30     printf("Rear = %d\n", queue[rear]);  
31  
32     return 0;  
33 }
```

Test Input

3
10
20
30

Output

Front = 10
Rear = 30

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📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int dequeue[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 void insertFront(int data)  
10 {  
11     if((front == 0 && rear == MAX - 1) || (front == rear + 1))  
12     {  
13         printf("Deque Overflow\n");  
14         return;  
15     }  
16     if (front == -1)  
17     {  
18         front = rear = 0;  
19     }  
20     else if (front == 0)  
21     {  
22         front = MAX - 1;  
23     }  
24     dequeue[front] = data;  
25 }  
26  
27 int main()  
28 {  
29     int n, i, x;  
30     scanf("%d", &n);  
31     for(i=0; i<n; i++)  
32     {  
33         scanf("%d", &x);  
34         insertFront(x);  
35     }  
36     printf("Front Element = %d\n", dequeue[front]);  
37     return 0;  
38 }
```

Test Input

```
3  
10  
20  
30
```

Output

Front Element = 30

⏱ Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x)  
4 {  
5     if(l>r)  
6         return -1;  
7     int mid = l + (r - l) / 2;  
8  
9     if(arr[mid] == x)  
10        return mid;  
11  
12    if (arr[mid] > x)  
13        return binarySearch(arr, l, mid - 1, x);  
14    return binarySearch(arr, mid + 1, r, x);  
15 }  
16 int main()  
17 {  
18     int n, i, x;  
19  
20     scanf("%d", &n);  
21     int arr[n];  
22  
23     for(i=0; i<n; i++)  
24         scanf("%d", &arr[i]);  
25     scanf("%d", &x);  
26  
27     int result = binarySearch(arr, 0, n-1, x);  
28  
29     if(result == -1)  
30         printf("Element not found");  
31     else  
32         printf("%d", result);  
33     return 0;  
34 }
```

Test Input

```
5  
10 20 30 40 50  
40
```

Output

3

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data)  
9 {  
10     if ((rear + 1) % MAX == front)  
11     {  
12         printf("Queue full\n");  
13         return;  
14     }  
15     if (front == -1)  
16         front = rear = 0;  
17     else  
18         rear = (rear + 1) % MAX;  
19     queue[rear] = data;  
20 }  
21 int dequeue()  
22 {  
23     if (front == -1)  
24         return -1;  
25     int data = queue[front];  
26  
27     if (front == rear)  
28     {  
29         front = rear = -1;  
30     }  
31     else  
32     {  
33         front = (front + 1) % MAX;  
34     }  
35     return data;  
36 }  
37 int main()  
38 {  
39     int n, i, x;  
40     scanf("%d", &n);  
41  
42     for(i=0; i<n; i++)  
43     {  
44         scanf("%d", &x);  
45         enqueue(x);  
46     }  
47     printf("%d\n", dequeue());  
48  
49     return 0;  
50 }
```

Test Input

3
10
20
30

Output

10

Pending manual review

Submitted

CO2_AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3  
4 int stack[100], top = -1;  
5  
6 void push(int x) {  
7     stack[++top] = x;  
8 }  
9  
10 int pop() {  
11     return stack[top--];  
12 }  
13  
14 int main() {  
15     char expression[] = "23+5*";  
16     int i = 0;  
17  
18     while (expression[i] != '\0') {  
19         if (isdigit(expression[i])) {  
20             push(expression[i] - '0');  
21         } else {  
22             int op1 = pop();  
23             int op2 = pop();  
24  
25             if (expression[i] == '+')  
26                 push(op1 + op2);  
27             else if (expression[i] == '-')  
28                 push(op2 - op1);  
29             else if (expression[i] == '*')  
30                 push(op1 * op2);  
31             else if (expression[i] == '/')  
32                 push(op2 / op1);  
33         }  
34         i++;  
35     }  
36     printf("%d", pop());  
37     return 0;  
38 }
```

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Test Input

stdin input (optional)

Output

25

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